AI PROJECT MANAGEMENT, INCEPTION AND PLANNING

MODULE AGENDA

Module Agenda

- Defining AI Project Vision
- Performing a Feasibility assessment & defining the Scope
- AI Project Metrics – Model, System, & Business
- Defining SMART Goals
- Key Stakeholders, Roles and Responsibilities in AI Projects
- Differences between Agile and Waterfall development
- Scaled Agile Framework (SAFe)
- Key Success Ingredients for AI Projects
- Risks and Management Strategies
- Budgeting and resource allocation

Recall the AI Project lifecycle

Our focus

Step 1. Problem Scoping
Define the business problem & objectives. Use the 4Ws (Who, What, Where, Why) to understand context & expected outcomes.

Step 2. Data Acquisition
Gather relevant data from internal and/or external sources. This includes structured & unstructured data collection through APIs, databases, sensors, etc.

Step 3. Data Exploration
Analyze the data to understand patterns, identify outliers, handle missing values, & perform feature engineering.

Step 4. Modeling
Develop & train AI models using selected algorithms. Fine-tune the model to improve accuracy & performance.

Step 5. Model Evaluation
Assess model performance using metrics such as accuracy, precision, recall, F1-score, etc. This helps ensure the model is reliable and generalizable.

Step 6. Deployment
Integrate the trained model into a production environment where it can be used by end users to generate predictions.

Step 7. Maintenance & Monitoring
Continuously monitor the model's performance and retrain it as needed to ensure it adapts to new data and remains effective over time.

DEFINING VISION, SCOPE & FEASIBILITY ASSESSMENT

Introduction to Managing AI Projects

- In this module, we will focus on project management fundamentals as they apply to AI projects.
- Successfully delivering an AI project is not just about algorithms and models; it requires careful planning, coordination, and alignment with business goals.
  - **Planning:** Define scope, success metrics, and timelines.
  - **Coordination:** Cross-functional communication and agile teams.
  - **Business Alignment:** Ensure ROI, KPIs, and stakeholder buy-in.

Defining AI Project Vision: Your "North Star"

- What does success look like? Clearly articulate the desired end-state.
- How will this AI initiative fundamentally improve or transform a part of the business?
- A strong vision guides all subsequent decisions and keeps the team focused.

Business Value Assessment: Why this Project?

- Before deploying resources to any AI project, the first step is to ensure strategic alignment and measurable impact by answering a key question: ***Why this project?***
- Leadership should clearly identify the following key pillars:

Quantify Expected Business Value: Estimate potential outcomes such as cost reduction (e.g., 15%), revenue growth (e.g., $1M annually), or time savings (e.g., 3,000 labor hours per year).

Align with Strategic Goals: Ensure the project directly supports key business objectives such as improving customer retention, increasing operational efficiency, or unlocking new revenue streams.

Define Clear Success Metrics: Establish well-defined Key Performance Indicators (KPIs) to track performance and ROI.

Feasibility Assessment: Can We Deliver This AI Project?

- Before moving forward, leadership must assess whether the AI project is feasible from technical, organizational, and data perspectives.

Technical Feasibility: Do we have the right tools and infrastructure (e.g., cloud, compute)?, Are the required AI models technically achievable given our use case?

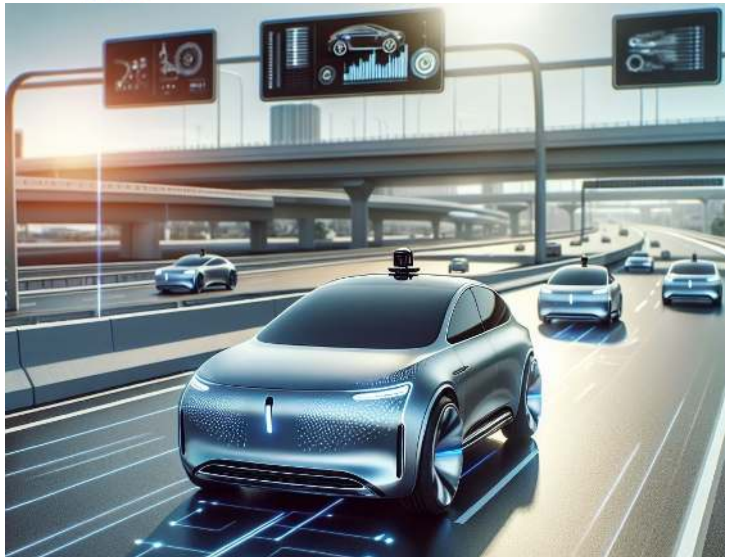
Organizational Readiness: Do we have skilled personnel (data scientists, ML engineers)?

Data Availability & Quality: Do we have access to the right data (historical, real-time, labeled)?, Is the data clean, complete, & enough to train a model?

Cost & Resource Constraints: Can we deliver the project within the available budget and timeframe?, Are there tools we can leverage to reduce development costs (e.g., AutoML, pre-trained models)?

Defining AI Project Scope

- It is critical to outline what the project will and will not deliver to manage expectations and avoid scope creep.
- Adopt a Minimum Viable Product (MVP) approach, focusing first on delivering core functionality that demonstrates value.
- As data understanding and insights evolve, revisit and validate the scope to ensure continued alignment with project goals and stakeholder needs.
- ***Example: Self-Driving Car MVP***



In Scope (MVP – Highway Driving)

Lane keeping on highways

Adaptive cruise control

Automatic lane changes in clear conditions

Operating in daylight and good weather

Out of Scope (Future Phases):

Urban driving (e.g., traffic lights, pedestrians, intersections)

Complex weather conditions (e.g., snow, heavy rain, fog)

Construction zones and unstructured environments

Nighttime driving with limited visibility

PRACTICE OPPORTUNITY

Practice Opportunity

Background

- UnitedHealth Group is one of the largest health insurance providers in North America. The company processes over 10 million claims yearly, with rising concerns about fraudulent activity, especially in dental and outpatient claims. Leadership wants to implement an AI model to automatically flag potentially fraudulent claims for human review, helping reduce unnecessary payouts and speed up claims processing.
- The data science team at UnitedHealth Group is small (2 data analysts and 1 software engineer), but the company has budgeted for hiring one external AI consultant to support the MVP phase. UnitedHealth Group already stores its claims data in a cloud database (BigQuery) and uses Google Cloud tools like AutoML and Vertex AI for analytics. Historical claim data includes structured fields such as: Type of procedure, Date of service, Claimed amount, Provider ID, and Region. The leadership team aims to launch a prototype in 6 months, starting with the most frequent and least complex claims.
- You are part of a team tasked with evaluating the feasibility of this AI project and defining its initial scope before the company invests resources.

Your Tasks

- Assess whether the project is technically and organizationally feasible.
- Define what the AI system should focus on in its first version (MVP) and what should be left for future phases.

Practice Opportunity Solution

Pillar	Feasible?	Reasoning
Technical Feasibility	Yes	UnitedHealth Group uses Google Cloud, BigQuery, and AutoML; sufficient for building a fraud detection prototype using structured data.
Organizational Readiness	Partially	The internal team is small, but support from an external AI consultant makes an MVP feasible. More hiring would be needed for full deployment.
Data Availability & Quality	Yes	The company has labeled historical structured data (procedure code, amount, approval outcome), which is clean and stored in a modern cloud system.
Cost & Resource Constraints	Yes	The project is scoped to 6 months, leveraging existing tools (AutoML) and an external consultant. This is realistic for a Phase 1 MVP.

Practice Opportunity Solution

In Scope (Phase 1 - MVP)	Out of Scope (Future Phases)
Focus on dental and outpatient claims	Complex claims such as surgical
Use only structured data (e.g., procedure code, cost)	No use of scanned documents or unstructured notes
Model flags suspicious claims for manual review only	No autonomous approvals or rejections of claims
Leverage AutoML for model development	No custom model building experimentation
Weekly batch predictions for claims	No real-time scoring or streaming data integration

AI PROJECT SUCCESS METRICS: MODEL, SYSTEM, & BUSINESS

AI Projects SMART Goals

- In AI projects, success metrics should follow the SMART criteria and be aligned with business, system, and model-level goals.

Specific

Measurable

Achievable

Relevant

Time-Bound

Example

- ***Goal:*** *Reduce average customer service response time by* ***40%*** *(from 5 minutes to 3 minutes) by* ***deploying an AI-powered chatbot*** *within* ***3 months****, while maintaining a* ***customer satisfaction score above 85%****.*

SMART Criterion	**Explanation**
Specific	Focused on reducing response time using a chatbot.
Measurable	40% reduction in response time and customer satisfaction score ≥ 85%.
Achievable	Based on industry benchmarks and current capacity.
Relevant	Aligns with business goals: improving user experience and reducing support costs.
Time-bound	Deadline set to 3 months.

Three Levels of AI Success Metrics: Model, System, Business

Model Quality Metrics

- Model Quality Metrics measure what your model produces.
- For traditional classification models: accuracy, Precision, Recall.
- For LLMs, could be coherence, instruction adherence, etc.

System Quality Metrics

- System Quality Metrics measure how well the model runs, such as deployment, reliability, latency, throughput, and resource efficiency.

Business Metrics

- These measure the real-world business impact of your AI system.
- Assesses productivity gains, cost savings, and customer experience improvements.
- Helps justify AI investments and prioritize high-impact use cases.

Success Metrics Examples

Business: Increase customer retention by 15% in the next 6 months using AI-driven product recommendations, while maintaining customer satisfaction above 90%.

Model: Achieve a minimum F1 score of 0.92 on the test dataset within 4 weeks of training, using cross-validation and hyperparameter tuning.

System: Maintain 99.9% uptime and ensure API response latency stays under 250ms for 95% of user requests over the next 3 months post-deployment.

PRACTICE OPPORTUNITY

STEMPLICITY

Practice Opportunity

- A bank wants to use AI to predict which customers are likely to miss their monthly loan payments. The goal is to help customer service teams proactively reach out and offer help before a payment is missed.
- Your team has been tasked to build the AI system, and you’ve been asked to define how success will be measured at three levels: **model, system, and business**. List 3 models, business, and system metrics in SMART goal format.

Practice Opportunity Solution

Category	Metric	SMART Goal
Model	Correct Prediction Rate	Achieve at least 85% accuracy in predicting missed payments within 3 months.
	False Positive Rate	Keep incorrect warnings under 10% during the pilot phase.
	False Negative Rate	Identify at least 90% of truly risky customers by the end of Q2.
System	Response Time	Ensure results are returned in under 2 seconds for 95% of users by launch.
	System Uptime	Maintain 99.5% uptime during business hours in the first 60 days.
	Scalability	Support 1000 predictions/hour without slowdown by end of first month.
Business	Missed Payment Reduction	Reduce missed loan payments by 12% within 3 months of system use.
	Proactive Outreach Rate	Increase early outreach calls by 25% in the first month after deployment.
	Cost Savings	Achieve a 10% reduction in payment collection costs within 6 months.

KEY AI PROJECTS STAKEHOLDERS AND ROLES

Key Roles in AI Projects

1. Business & Product Roles

2. Data Science & AI Roles

3. Development & Operations Roles

4. Data Engineering Roles

Summary

Role	Key Responsibilities
AI Product Manager	Bridges technical and business teams; defines vision, translates problems, sets metrics
Industry Domain Expert	Provides subject matter expertise, validates data and features
Project Manager	Coordinates project scope, timelines, communication, and risk
AI Researcher	Explores novel models and learning techniques
Data Scientist	Analyzes data, builds models, extracts insights
Machine Learning Engineer	Builds scalable ML pipelines, deploys models
Responsible AI / Ethics Lead	Audits models for fairness, ethics, and bias
UX Designer	Designs intuitive AI interfaces and feedback loops
Software Engineer	Develops applications integrating AI models
Solution Architect	Designs system architecture and integration plan
MLOps Architect	Manages deployment, monitoring, and updates
Data Engineer	Builds ETL pipelines, structures and cleans data
Data Architect	Designs data infrastructure, governance policies

1. Business and Product Roles: **AI Product Manager**

Role Description:
- Bridges the gap between the technical team and business stakeholders. They must understand AI-specific trade-offs (e.g., accuracy vs interpretability) and feasibility.

Key Responsibilities:
- Define the AI product vision & roadmap.
- Translate business problems into machine learning objectives.
- Work closely with data scientists, engineers, and UX teams.
- Define success metrics (e.g., model & business KPIs).
- Prioritize features based on user feedback, risk, and impact.

Why It's Critical:
- AI initiatives often fail due to poor alignment with business goals; this role ensures the AI solution solves a real, measurable problem.

Lead Product Manager, AI

Lattice - San Francisco, CA
$179,000 - $223,500 a year

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- Strategize closely with senior leadership and cross-functional teams to shape the future of Lattice's AI strategy and roadmap.
- Spearhead the development of multiple AI-powered features and agentic systems leveraging LLMs and/or ML technologies.
- Conduct thorough market analysis, user research, and data gathering to identify opportunities where AI can solve customer problems and deliver exceptional value.
- Define and refine product wireframes, scope, requirements, and detailed specifications for AI features, ensuring they are grounded in both technical feasibility and user needs.
- Establish and monitor key AI product KPIs and metrics, evaluating performance to guide continuous improvement and ensure AI experiences meet quality, safety, and ethical standards.
- Drive cross-functional collaboration with design, engineering, AI/ML, user research, product marketing, security, and legal teams to deliver sophisticated AI solutions that align with business goals.
- Maintain a deep understanding of AI/ML technology trends and competitive landscape, positioning Lattice as a thought leader in AI-powered HR solutions.

What You Will Bring to the Table

- Minimum 7 years of progressive experience in product management, with specific expertise in AI/ML product development, preferably in B2B SaaS environments.
- A track record of success in leading AI product development from 0-to-1, with demonstrated business impact and customer adoption.
- Strong leadership skills, with experience mentoring other product managers and guiding cross-functional teams through complex AI initiatives.
- Deep understanding of AI/ML technologies, including generative AI, LLMs, and how they can be applied to workflow-heavy products.
- Strategic thinking with the ability to navigate complex tradeoffs across UX, data availability, AI reliability, technical complexity, and business priorities.
- Excellent communication skills, adept at articulating complex AI concepts to diverse audiences and influencing at all levels of the organization.
- Robust analytical skills, skilled in using data and metrics for decision-making.
- Entrepreneurial mindset, thriving in ambiguity and taking ownership in a dynamic environment.

https://www.indeed.com/cmp/Lattice-6/jobs?jk=5e803e3e68aeb306&start=0

1. Business and Product Roles: **Industry Domain Expert**

Role Summary:
- Provides the real-world context and subject matter expertise.

Key Responsibilities:
- Explain domain-specific details to the technical team (e.g., healthcare codes, financial regulations).
- Validate data relevance and feature engineering.
- Provide data labelling criteria & analyze model outputs in a business context.

Why It's Critical:
- AI models trained without domain context can make misleading or even dangerous predictions (e.g., in finance, healthcare). Domain expertise ensures relevance and accuracy.

1. Business and Product Roles: **Project Manager**

Role Summary:

- Coordinates all moving parts of an AI project to ensure it is delivered on time, within scope, and on budget.

Key Responsibilities:

- Create and manage the project plan.
- Track milestones and deliverables.
- Facilitate communication across cross-functional teams.
- Mitigate risks, manage dependencies, and ensure resource availability and allocation.

Why It’s Critical:

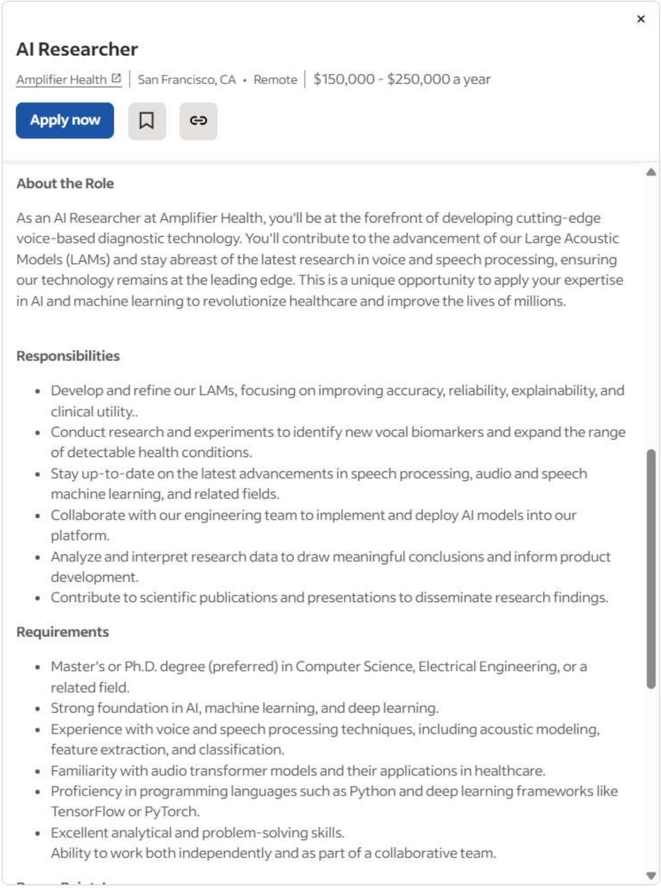
- AI projects often span across departments and involve iterative development. A project manager keeps the team focused and aligned.

2. Data Science & AI Roles: **AI Researcher**

Role Summary:
- Pushes the boundaries of AI and ML theory, often working on novel models, algorithms, or use cases.

Key Responsibilities:
- Develop new learning architectures (e.g., transformers).
- Publish papers, run experiments, and stay ahead of the curve.
- Explore cutting-edge techniques (e.g., self-supervised learning, few-shot learning).

Why It’s Critical:
- Enables innovation and differentiation, especially in R&D-heavy companies AI/GenAI work.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AI Researcher

Amplifier Health | San Francisco, CA • Remote | $150,000 - $250,000 a year

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About the Role

As an AI Researcher at Amplifier Health, you'll be at the forefront of developing cutting-edge voice-based diagnostic technology. You'll contribute to the advancement of our Large Acoustic Models (LAMs) and stay abreast of the latest research in voice and speech processing, ensuring our technology remains at the leading edge. This is a unique opportunity to apply your expertise in AI and machine learning to revolutionize healthcare and improve the lives of millions.

Responsibilities

- Develop and refine our LAMs, focusing on improving accuracy, reliability, explainability, and clinical utility..
- Conduct research and experiments to identify new vocal biomarkers and expand the range of detectable health conditions.
- Stay up-to-date on the latest advancements in speech processing, audio and speech machine learning, and related fields.
- Collaborate with our engineering team to implement and deploy AI models into our platform.
- Analyze and interpret research data to draw meaningful conclusions and inform product development.
- Contribute to scientific publications and presentations to disseminate research findings.

Requirements

- Master's or Ph.D. degree (preferred) in Computer Science, Electrical Engineering, or a related field.
- Strong foundation in AI, machine learning, and deep learning.
- Experience with voice and speech processing techniques, including acoustic modeling, feature extraction, and classification.
- Familiarity with audio transformer models and their applications in healthcare.
- Proficiency in programming languages such as Python and deep learning frameworks like TensorFlow or PyTorch.
- Excellent analytical and problem-solving skills.
  Ability to work both independently and as part of a collaborative team.

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2. Data Science & AI Roles: **Data Scientist**

Role Summary:

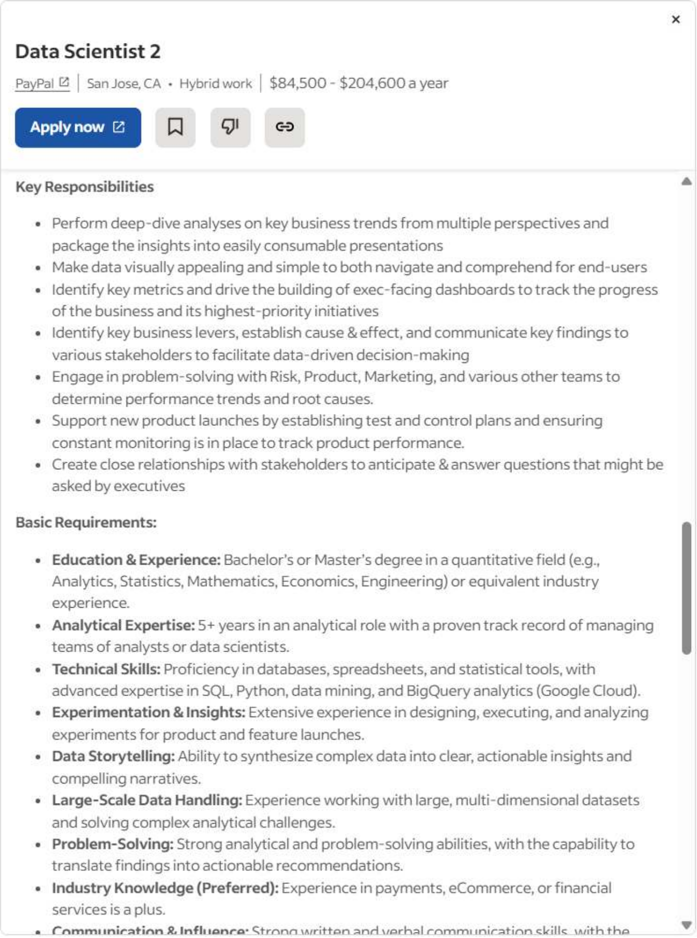
- Solves complex business problems using data. Uses statistical analysis and machine learning to derive insights and build predictive models.

Key Responsibilities:

- Analyze structured and unstructured data.
- Develop and test ML models.
- Perform feature engineering.
- Communicate insights to stakeholders.

Why It's Critical:

- Data scientists turn raw data into business value, guiding decisions and automating intelligent processes.



Data Scientist 2

PayPal | San Jose, CA • Hybrid work | $84,500 - $204,600 a year

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Key Responsibilities

- Perform deep-dive analyses on key business trends from multiple perspectives and package the insights into easily consumable presentations
- Make data visually appealing and simple to both navigate and comprehend for end-users
- Identify key metrics and drive the building of exec-facing dashboards to track the progress of the business and its highest-priority initiatives
- Identify key business levers, establish cause & effect, and communicate key findings to various stakeholders to facilitate data-driven decision-making
- Engage in problem-solving with Risk, Product, Marketing, and various other teams to determine performance trends and root causes.
- Support new product launches by establishing test and control plans and ensuring constant monitoring is in place to track product performance.
- Create close relationships with stakeholders to anticipate & answer questions that might be asked by executives

Basic Requirements:

- **Education & Experience:** Bachelor's or Master's degree in a quantitative field (e.g., Analytics, Statistics, Mathematics, Economics, Engineering) or equivalent industry experience.
- **Analytical Expertise:** 5+ years in an analytical role with a proven track record of managing teams of analysts or data scientists.
- **Technical Skills:** Proficiency in databases, spreadsheets, and statistical tools, with advanced expertise in SQL, Python, data mining, and BigQuery analytics (Google Cloud).
- **Experimentation & Insights:** Extensive experience in designing, executing, and analyzing experiments for product and feature launches.
- **Data Storytelling:** Ability to synthesize complex data into clear, actionable insights and compelling narratives.
- **Large-Scale Data Handling:** Experience working with large, multi-dimensional datasets and solving complex analytical challenges.
- **Problem-Solving:** Strong analytical and problem-solving abilities, with the capability to translate findings into actionable recommendations.
- **Industry Knowledge (Preferred):** Experience in payments, eCommerce, or financial services is a plus.

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2. Data Science & AI Roles: **Machine Learning Engineer**

Role Summary:

- Builds, tests, and deploys scalable machine learning pipelines. Often bridges the gap between data science and engineering.

Key Responsibilities:

- Design ML pipelines for production.
- Tune models for performance and scalability.
- Integrate models into applications.
- Collaborate with software and MLOps teams.

Why It’s Critical:

- ML engineers make AI real by taking models out of notebooks and into production.

2. Data Science & AI Roles: **Responsible AI / Ethics Lead**

Role Summary:

- Ensures AI systems are ethical, fair, transparent, and inclusive. Evaluates models for unintended harm or bias.

Key Responsibilities:

- Audit data and models for bias and discrimination.
- Enforce fairness and explainability standards.
- Ensure ethical practices in data labelling and use.

Why It’s Critical:

- Helps prevent ethical issues, reputational damage, and regulatory violations in high-stakes domains.

2. Data Science & AI Roles: **UX Designer**

Role Summary:

- Designs intuitive interfaces for AI systems that enhance user trust, interpretability, and usability.

Key Responsibilities:

- Create user journeys for interacting with AI outputs.
- Simplify complex data/model results for end users.
- Design feedback loops (e.g., for human-in-the-loop systems).

Why It’s Critical:

- If users can’t understand or interact with the AI, they won’t use it, regardless of how powerful it is.

3. Develpment and Operations Roles: **Software Engineer**

Role Summary:

- Builds applications, APIs, and tools that integrate AI models into real-world products and user experiences.

Key Responsibilities:

- Develop front-end and back-end systems.
- Connect AI models to user interfaces and data sources.
- Ensure code quality, scalability, and testing.
- Work closely with ML engineers and UX designers.

- **Why It’s Critical:**
  Even the best AI model is useless if it’s not integrated into a product. Software engineers turn models into usable tools.

3. Develpment and Operations Roles: **Solution Architect**

Role Summary:

- Designs the technical blueprint for how the AI system will be built, deployed, and maintained across the stack.

Key Responsibilities:

- Define system architecture (data flow, storage, model interfaces).
- Choose the right tools, platforms, and design patterns.
- Ensure scalability, security, and interoperability.

Why It's Critical:

- Poor architecture means slow systems, high costs, and rework. A good architect sets the foundation for success.

3. Develpment and Operations Roles: **MLOps Architect**

Role Summary:

- MLOps engineers ensure that AI models don’t just live in notebooks; they make sure models are deployed, monitored, and continuously updated in production.

Key Responsibilities:

- Automate model deployment (CI/CD pipelines).
- Monitor model performance (accuracy, latency, drift).
- Manage model versioning and rollback strategies.
- Collaborate with data scientists and IT teams.

Why It’s Critical:

- AI is not just about building models; it’s about running them reliably in the real world. MLOps ensures smooth, scalable operations.

4. Data Engineering Roles: **Data Engineer**

Role Summary:

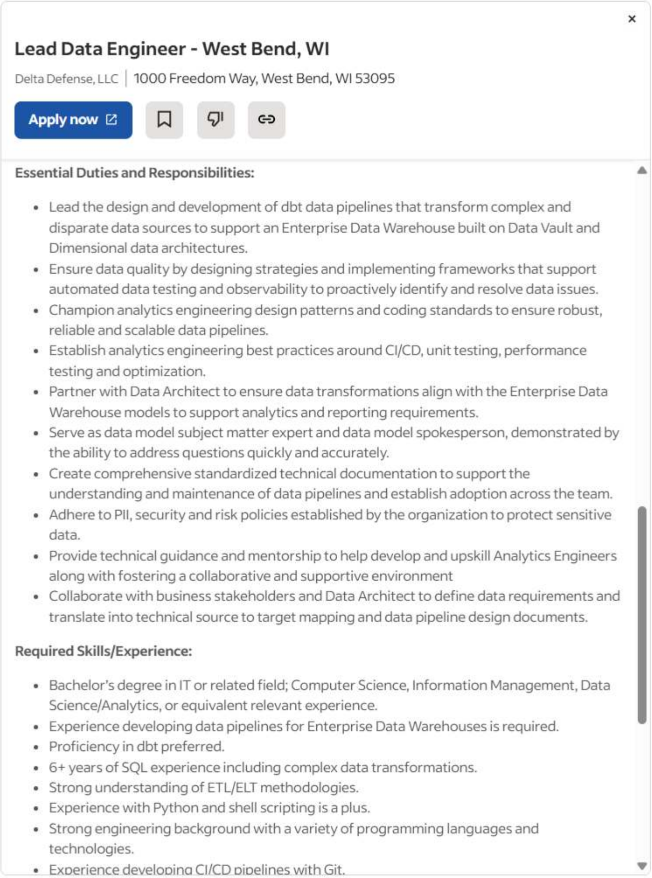
- Builds and maintains the data pipelines that feed machine learning models and dashboards.

Key Responsibilities:

- Design and implement ETL/ELT pipelines.
- Clean, transform, and structure raw data.
- Connect and integrate various data sources (APIs, databases, files).
- Collaborate with data scientists to provide model-ready data.

Why It's Critical:

- No data = no AI. Data engineers are the backbone of the AI workflow, ensuring that clean, reliable data is always available.



Lead Data Engineer - West Bend, WI

Delta Defense, LLC | 1000 Freedom Way, West Bend, WI 53095

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Essential Duties and Responsibilities:

- Lead the design and development of dbt data pipelines that transform complex and disparate data sources to support an Enterprise Data Warehouse built on Data Vault and Dimensional data architectures.
- Ensure data quality by designing strategies and implementing frameworks that support automated data testing and observability to proactively identify and resolve data issues.
- Champion analytics engineering design patterns and coding standards to ensure robust, reliable and scalable data pipelines.
- Establish analytics engineering best practices around CI/CD, unit testing, performance testing and optimization.
- Partner with Data Architect to ensure data transformations align with the Enterprise Data Warehouse models to support analytics and reporting requirements.
- Serve as data model subject matter expert and data model spokesperson, demonstrated by the ability to address questions quickly and accurately.
- Create comprehensive standardized technical documentation to support the understanding and maintenance of data pipelines and establish adoption across the team.
- Adhere to PII, security and risk policies established by the organization to protect sensitive data.
- Provide technical guidance and mentorship to help develop and upskill Analytics Engineers along with fostering a collaborative and supportive environment
- Collaborate with business stakeholders and Data Architect to define data requirements and translate into technical source to target mapping and data pipeline design documents.

Required Skills/Experience:

- Bachelor's degree in IT or related field; Computer Science, Information Management, Data Science/Analytics, or equivalent relevant experience.
- Experience developing data pipelines for Enterprise Data Warehouses is required.
- Proficiency in dbt preferred.
- 6+ years of SQL experience including complex data transformations.
- Strong understanding of ETL/ELT methodologies.
- Experience with Python and shell scripting is a plus.
- Strong engineering background with a variety of programming languages and technologies.
- Experience developing CI/CD pipelines with Git.

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4. Data Engineering Roles: **Data Architect**

Role Summary:
- Designs the overall data infrastructure and flow; ensuring that data is scalable, secure, and efficient to access.

Key Responsibilities:
- Define data models, schemas, and architecture (e.g., lake vs warehouse).
- Choose the right tools and platforms (Snowflake, BigQuery, etc.).
- Ensure data storage is scalable and cost-effective.
- Set best practices for data access, retention, and governance.

Why It’s Critical:
- Without a sound structure, data becomes siloed, slow, or expensive to manage. The architect sets the foundation for all data-driven work.

PRACTICE OPPORTUNITY

STEMPLICITY

Practice Opportunity

- Your company is launching an AI-powered customer support chatbot to reduce wait times and improve customer satisfaction. The chatbot should be able to understand customer inquiries, provide instant answers, and escalate to a human agent when needed. You are part of the project team, and your first task is to map responsibilities to the right stakeholders based on the roles in AI projects.
- Below is a list of tasks involved in the project. Assign one appropriate role (from the summary table) to each task based on its key responsibilities.

Task	Assigned Role (Student to fill in)
Define the overall AI vision and align it with the business goals	
Design and train the chatbot's natural language processing model	
Validate the customer complaint categories with real examples	
Set up the cloud infrastructure to store and process chatbot data	
Monitor the chatbot's accuracy and response time in production	
Ensure fairness and avoid bias in chatbot responses	
Build the data pipelines to collect, clean, and store customer chats	
Connect the chatbot to the company's CRM system	
Design how users interact with the chatbot and receive feedback	
Coordinate the project timeline, sprints, and risk mitigation plans	

Practice Opportunity Solution

Task	Assigned Role
Define the overall AI vision and align it with the business goals	**AI Product Manager**
Design and train the chatbot’s natural language processing model	**AI Researcher** or **Data Scientist**
Validate the customer complaint categories with real examples	**Industry Domain Expert**
Set up the cloud infrastructure to store and process chatbot data	**Solution Architect**
Monitor the chatbot’s accuracy and response time in production	**MLOps Architect**
Ensure fairness and avoid bias in chatbot responses	**Responsible AI / Ethics Lead**
Build the data pipelines to collect, clean, and store customer chats	**Data Engineer**
Connect the chatbot to the company’s CRM system	**Software Engineer**
Design how users interact with the chatbot and receive feedback	**UX Designer**
Coordinate the project timeline, sprints, and risk mitigation plans	**Project Manager**